

Ideation Phase

Brainstorm & Idea Prioritization

Attribute	Details
Date	08 March 2026
Team ID	
Project Name	VoltEdge: Electric Vehicle Data Analytics Dashboard
Maximum Marks	4 Marks
Team Leader	Sanskriti Roman
Team Members	Lavani Sardar, Sanskriti Sawant, Sayali Firake

Step-1: Team Gathering, Collaboration and Selection

The team convened to discuss the critical pain points in the current Indian EV ecosystem. We identified three primary gaps: lack of centralized charging data, difficulty in comparing battery efficiency across brands, and "range anxiety" among potential buyers.

Finalized Problem Statement:

"Develop an Electric Vehicle Analytics Dashboard that analyzes EV charging patterns, battery performance, and model efficiency. The system leverages PostgreSQL for scalable data storage, Tableau for high-fidelity interactive visualizations, and Flask for seamless web-based integration."

Step-2: Brainstorm, Idea Listing and Grouping

During the brainstorming session, we utilized a "Mind Mapping" technique to categorize our ideas based on stakeholder needs.

- Category 1: Infrastructure Insights (For Planners): Charging station density by region, AC vs. DC charger distribution, and peak usage hours.
- Category 2: Vehicle Performance (For Fleet Managers): Efficiency ratings (\$Wh/km\$), PowerTrain analysis, and Top Speed vs. Range comparisons.
- Category 3: Consumer Decision Support (For Buyers): Price-to-Range ratio, Brand popularity, and Body Style (SUV vs. Sedan) availability.
- Category 4: Environmental Impact: CO2 savings estimation compared to internal combustion engines (ICE).

Step-3: Idea Prioritization

The team used a Feasibility vs. Impact Matrix to select the features for the first release (Sprint-1). We prioritized features that could be built using our PostgreSQL dataset and rendered effectively in Tableau.

Key Feature Importance for the Dashboard

Feature Name	Description	Priority	Impact
Regional Charging Density	Mapping charging stations across Noida, NDMC, etc.	High	Essential for city infrastructure planning (Aarav's Story).
EV Efficiency Index	Ranking brands based on \$Wh/km\$ and battery performance.	High	Directly addresses fleet operational costs (Meera's Story).
Market Brand Analysis	A bubble chart/bar chart showing brand market share.	High	Helps consumers understand market trends and reliability.

Feature Name	Description	Priority	Impact
Price vs. Range Scatter	Identifying the "Value for Money" vehicle models.	Medium	High impact for buyers but requires precise pricing data.
Predictive Battery Health	Using ML to predict battery degradation over time.	Low	High technical complexity; reserved for Future Scope.

Final Selection for Phase-II

Based on this prioritization, we decided to focus our Tableau development on 11 high-impact visualizations, specifically focusing on the Charging Station Distribution and the EV Brand Efficiency Ranking, as these provide the most immediate value to our identified users.